

React Router

React Router is a standard **routing library** for React that enables **navigation** between different pages or views in a React app — **without reloading the browser**.

It is essential for building **Single Page Applications (SPA)**, where you don't want the whole page to reload when navigating between different parts of the app.

Core Components of React Router

1. <BrowserRouter>

- Wraps your entire app.
- Enables routing using browser's URL.

```
import { BrowserRouter } from 'react-router-dom';
```

```
<BrowserRouter>
```

```
  { /* Your app here */ }
```

```
</BrowserRouter>
```

2. <Routes> and <Route>

- Define which component should show for which path.

```
<Routes>
```

```
<Route path="/" element= {<Home />} />
```

```
<Route path="/about" element= {<About />} />
```

```
</Routes>
```

3. <Link>

- Used instead of <a> to navigate between routes **without refreshing the page**.

```
<Link to="/about">Go to About</Link>
```

4. useNavigate()

- Hook used for **programmatic navigation** (e.g. go to /login after form submit).

```
const navigate = useNavigate();
```

```
navigate('/login');
```

Benefits of Using React Router

- Fast navigation (no page reload)
- Better user experience (feels like a native app)
- Cleaner URL management
- Support for dynamic routing (e.g., /user/:id)
- Useful for auth and protected routes

Example :

1.npx create-react-app reactrouter-app

2.src/

└─ components/

| └─ Home.js

| └─ About.js

└─ App.js

3. src/components/Home.js

```
import React from 'react';
```

```
function Home() {  
  return <h2>Home Page</h2>;  
}
```

```
export default Home;
```

4.src/components/About.js

```
import React from 'react';
```

```
function About() {  
  return <h2>About Page</h2>;  
}
```

```
export default About;
```

5.src/App.js

```
import React from 'react';  
import { BrowserRouter as Router, Routes, Route, Link } from 'react-router-dom';  
import Home from './components/Home';  
import About from './components/About';
```

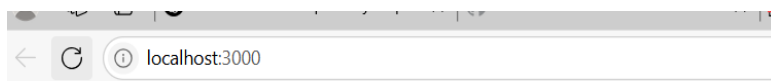
```
function App() {  
  return (  
    <Router>
```

```
<nav style={{ marginBottom: '20px' }}>
  <Link to="/" style={{ marginRight: '10px' }}>Home</Link>
  <Link to="/about">About</Link>
</nav>

<Routes>
  <Route path="/" element={<Home />} />
  <Route path="/about" element={<About />} />
</Routes>
</Router>
);
}
export default App;
```

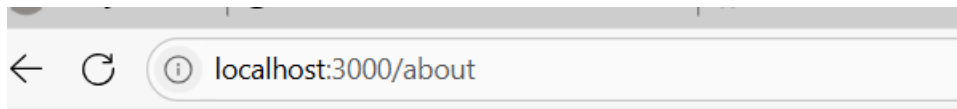
6.Make sure react-router-dom is installed: `npm install react-router-dom`

7.npm start



Home About

Home Page



[Home](#) [About](#)

About Page